Introducing CicatrixVision (CiVi) Dataset: First steps into AI assisted Chronic Wound Assessment

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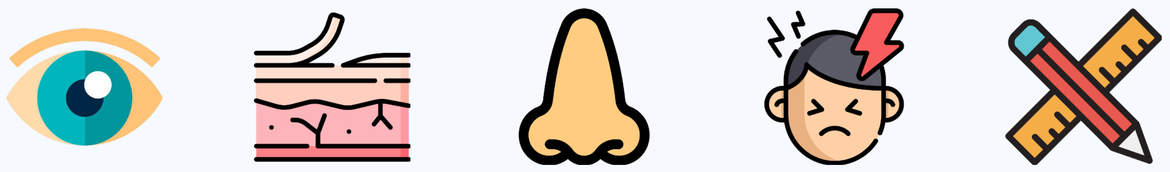
Chronic Wound Healing

Chronic Wounds



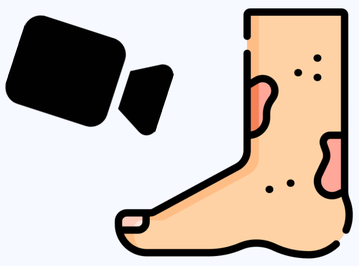
Do not heal in expected time

Current Assesment Parameters



Subjective and difficult to standardize

Objective



Estimate tissue composition using AI

Motivation

Provide CiVi: an open source dataset with high-quality annotations for wound tissue types

CicatrixVision (CiVi) Dataset

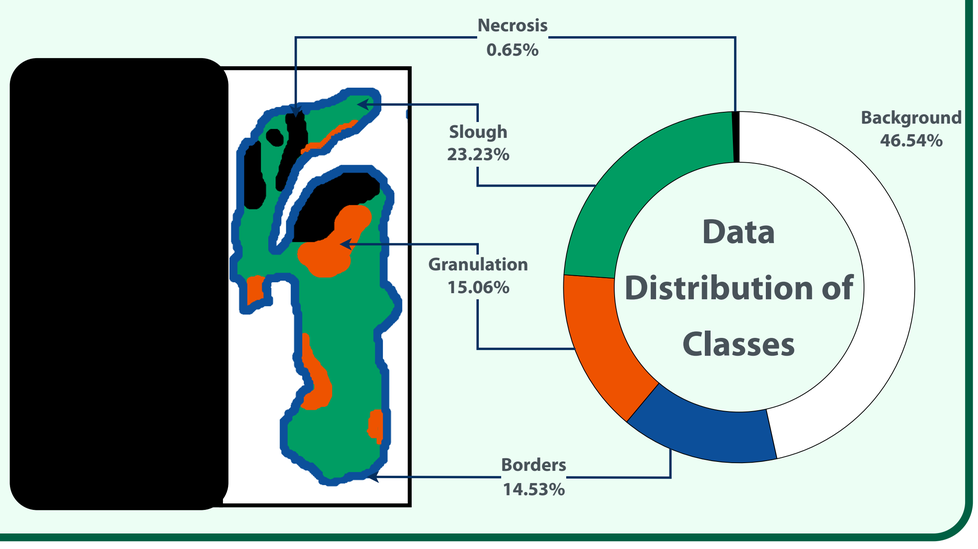
6500 High Quality Images — UKR Universitätsklinikum Regensburg

→ **Filtered by:**
- Year - 2014
- Quality of Image
- Wound Types

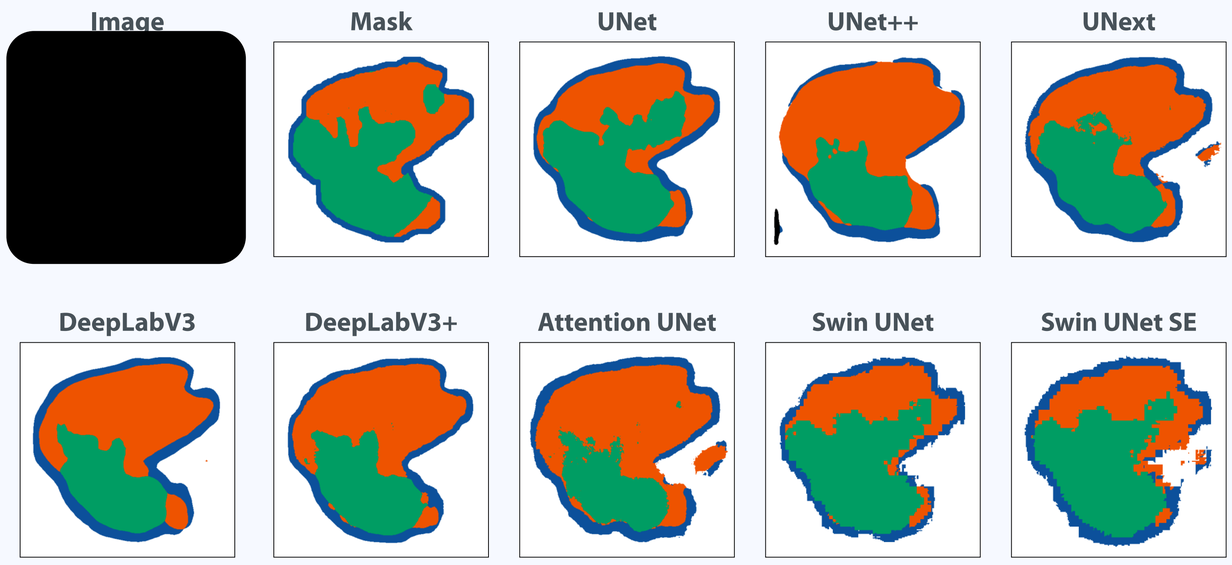
→ **Segment and crop using WSNet[3]**

- 66 Patients
- 3 Tissue Classes
- 721 Wound Images
- Label with LabelStudio

5 Types of Wounds
- Venous Ulcers
- Arterial Ulcers
- Vasculitis
- Graft-versus-Host Disease
- Pyoderma Gangrenosum



Benchmark Tests and Results



Model	Backbone	Loss Function	Accuracy	IoU	Dice
UNet	**EfficientNetB0**	**RCE**	**0.82**	**0.67**	**0.80**
UNet++	EfficientNetB2	DL	0.78	0.52	0.66
UNext	Baseline	RCE	0.76	0.63	0.77
DeepLabV3	EfficientNetB2	CE	0.81	0.60	0.73
DeepLabV3+	EfficientNetB0	CE	0.77	0.62	0.75
Attention UNet	Baseline	RCE	0.64	0.51	0.66
Swin UNet	Baseline	FL	0.75	0.61	0.75
Swin UNet	With SE	FL	0.73	0.61	0.75

Conclusion & Future Work

- Design a Computer Vision training architecture using spatial knowledge about the structure of the wound images.
- Utilise the vast amount of unannotated images in a Semi-Supervised Learning pipeline.
- Understand the relationship between wound tissue composition and wound healing in chronic wounds.

References

[1] Izadi, K. (2005). Chronic wounds. ***Clinics in plastic surgery***.

[2] Bowers, S. (2020). Chronic wounds: evaluation and management. ***American family physician***.

[3] Oota, S.(2023). WSNet: towards an effective method for wound image segmentation. In ***Proceedings of the IEEE/CVF WACV.***

[4] Ramachandram, D.(2022). Fully automated wound tissue segmentation using deep learning on mobile devices: Cohort study. ***JMIR mHealth and uHealth***.

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